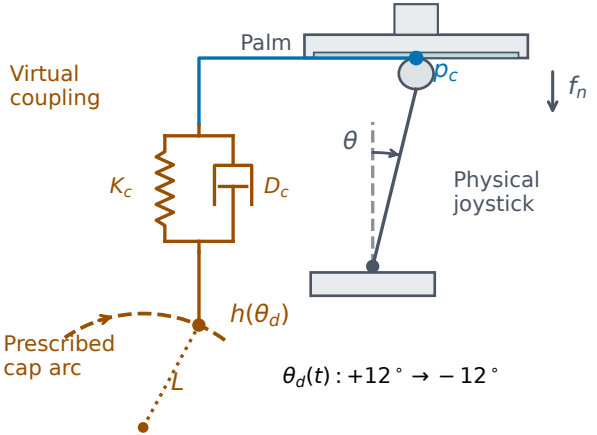
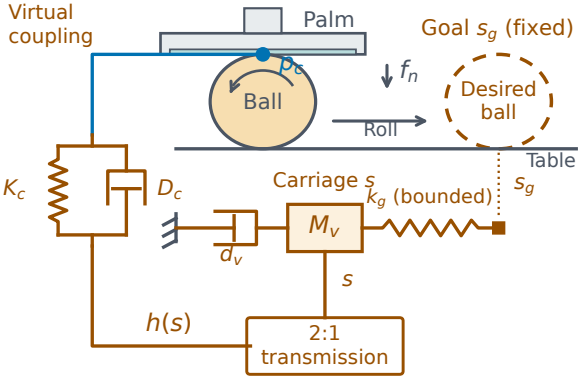
(a) Joystick: arc-guided spring–damper



Normal preload: K_n, D_n; orientation follows θ_d

EIT: define VMC attachment at contact

(b) Ball rolling: virtual carriage

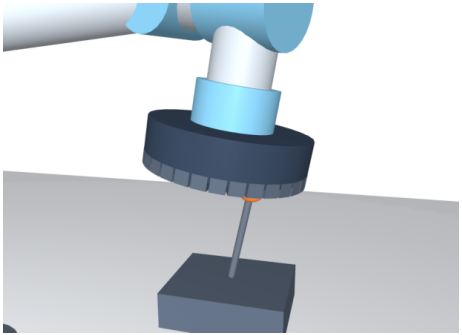


Normal preload: K_n, D_n; orientation: horizontal

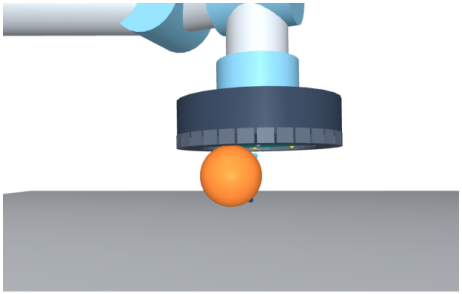
EIT: update attachment + correct progress

Rebase reference to preserve spring extension

(c) Joystick simulation



(d) Ball simulation



Amber: virtual components Blue dots / links: selected attachment Lower panels: saved MuJoCo states

$$f_c = P_t[K_c(h - p_c) + D_c(v_{\text{ref}} - J_c\dot{q})], \qquad \tau_{task} = J_c^T(f_c + f_n) + J_\omega^T m_R$$